

Parametric Form

- Find the Cartesian equation of the curves with the parametric equations given.
 - $x = 2t, y = t + 2$
 - $x = t, y = t^2$
 - $x = t, y = \frac{1}{t}$
 - $x = 2\cos\theta, y = 2\sin\theta, 0 \leq \theta \leq 2\pi$
 - $x = 2\cos\theta, y = 2\sin\theta, 0 \leq \theta \leq \pi$
 - $x = t + 3, y = t^2 - 5, t \geq 0$
 - $x = 2u - 2, y = 3u + 1, 1 \leq u \leq 3$
 - $x = v^3, y = 1 - v^2, -1 \leq v \leq 1$
 - $x = t + 2, y = t^2 - 1$
 - $x = \cos t, y = \sin t, 0 \leq t \leq 2\pi$
 - $x = 2t^2, y = 4t$
 - $x = 2\cos\theta, y = \sqrt{3}\sin\theta, 0 \leq \theta \leq 2\pi$
 - $x = 2\cos t, y = \sin t, 0 \leq t \leq \pi$
 - $x = \frac{2t}{1+t^2}, y = \frac{1-t^2}{1+t^2}$
- Two boats on a lake start sailing at the same time. Boat A moves on a course given by $x = \frac{t}{2}, y = t + 1$, while boat B moves on a course given by $x = t - 2, y = -2t + 9$, where t is the time elapsed in hours.
 - Find the Cartesian equation for the course of each boat. Show that the courses intersect at the point $(1, 3)$.
 - Do the boats collide? Justify your answer.
- Find the Cartesian equation for the parametric equations $x = \sec\theta, y = \tan\theta, -\frac{\pi}{2} < \theta < \frac{\pi}{2}$.
- Find the Cartesian equation for each of the following parabolas.
 - $x = 2t, y = 2t^2$
 - $x = 12t, y = 6t^2$
 - $x = t + 2, y = t^2$
 - $x = 4t, y = -2t^2$
 - $x = \sqrt{t}, y = t + 1$
- The parametric equations $x = 2t - 3$ and $6t - 5$ represent a line.
 - Find the points A and B with parameters $t = 0$ and $t = 1$, and hence find the gradient of the line.
 - Find the value of t that makes $x = 0$, and hence find the y -intercept.
 - Check your answers by eliminating t to form a Cartesian equation.
 - Repeat these steps for the lines:
 - $x = 2t - 3$ and $y = 3t - 2$
 - $x = at + b$ and $y = ct + d$
- Show that $x = a + r\cos\theta$ and $y = b + r\sin\theta$ define a circle with centre (a, b) and radius r .
 - Hence sketch a graph of the curve $x = 1 + 2\cos\theta, y = -3 + 2\sin\theta$.
- Show by elimination that $x = \frac{t^2-1}{t^2+1}$ and $y = \frac{2t}{t^2+1}$ almost represent the unit circle $x^2 + y^2 = 1$. What point is missing?
- Different parametric representations may result in the same Cartesian equation. The graphical representation, however, may be different.
 - Find the Cartesian equation of the curve $x = 2 - t, y = t - 1$ and sketch its graph.
 - Find the Cartesian equation of the curve $(\sin^2 t, \cos^2 t)$. Explain why $0 \leq x \leq 1$ and $0 \leq y \leq 1$ and sketch a graph of the curve.
 - Find the Cartesian equation of the curve $x = 4 - t^2, y = t^2 - 3$. Explain why $x \leq 4$ and $y \geq -3$ and sketch a graph of the curve.
- Find the Cartesian equation of the curve $x = 3 + r\cos\theta, y = -2 + r\sin\theta$, and describe it geometrically if:
 - r is constant and θ is variable.
 - θ is constant and r is variable.

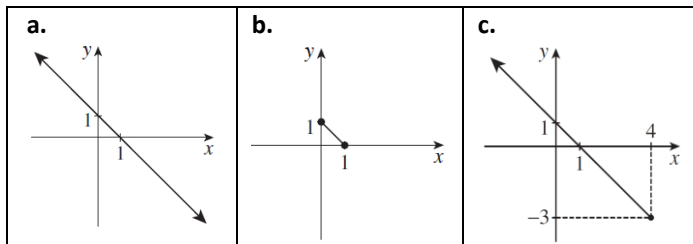
10. a. Show by elimination that $x = \frac{2t+1}{2t^2+2t+1}$ and $y = \frac{2t^2+2t}{2t^2+2t+1}$ almost represent the unit circle $x^2 + y^2 = 1$. What point is missing?
 b. Is this map from parameters to points one-to-one or many-to-one?
11. After finding the Cartesian equation, sketch the curve whose parametric equations are $x = \frac{1}{2}(2^t + 2^{-t})$ and $y = \frac{1}{2}(2^t - 2^{-t})$.
12. Sketch the following Parametric Equations ($c > 0$)

a. $x = c(t^2 + 1), y = \frac{c}{t^2 + 1}$
 b. $x = c(1 - t^2), y = \frac{c}{1 - t^2}$
 c. $x = c\sqrt{1 - t^2}, y = \frac{c}{\sqrt{1 - t^2}}$

d. $x = c \sin t, y = \frac{c}{\sin t}$
 e. $x = c(t^2 - 1), y = \frac{c}{t^2 - 1}$
 f. $x = c\sqrt{t^2 - 1}, y = \frac{c}{\sqrt{t^2 - 1}}$

Answer

1. a. $2y = x + 4$; b. $y = x^2$; c. $y = \frac{1}{x}, x \neq 0$; d. $x^2 + y^2 = 4, -2 \leq x \leq 2$; e. $y = \sqrt{4 - x^2}, -2 \leq x \leq 2$; f. $y = x^2 - 6x + 4, x \geq 3$; g. $2y = 3x + 8, 0 \leq x \leq 4$; h. $y = 1 - x^{\frac{2}{3}}, -1 \leq x \leq 1$; i. $y = x^2 - 4x + 3$; j. $y = x, -1 \leq x \leq 1$; k. $y^2 = 8x, x \geq 0$; l. $3x^2 + 4y^2 = 12, -2 \leq x \leq 2$; m. $y = \frac{1}{2}\sqrt{4 - x^2}, -2 \leq x \leq 2$; n. $x^2 + y^2 = 1, 1 \leq x \leq 1$
2. a. $y = 2x + 1, y = -2x + 5$; b. No. Boat A is at (1, 3) when $t = 2$, Boat B is at (1, 3) when $t = 3$.
3. $x^2 - y^2 = 1, x \geq 1$
4. a. $y = \frac{x^2}{2}$; b. $y = \frac{x^2}{24}$; c. $y = (x - 2)^2$; d. $y = -\frac{x^2}{8}$; e. $y = x^2 + 1, x \geq 0$
5. a. i. $A = (-3, -5), B = (-1, 1)$, gradient = 3; ii. When $x = 0, t = 1\frac{1}{2}$, so $y = 4$; iii. $y = 3x + 4$;
 b. i. $y = \frac{3}{2}x + \frac{5}{2}$; ii. $y = \frac{cx}{a} + \frac{ad - bc}{a}$
6. b. center (1, -3) $r = 2$
7. The point (1, 0) is missing, because when $y = 0, t = 0$, so $x = -1$
- 8.



9. a. $(x - 3)^2 + (y + 2)^2 = r^2$, circle with centre (3, -2) and radius r ;
 b. $y = x \tan \theta - (3 \tan \theta + 2)$, straight line with gradient $\tan \theta$
10. a. The point (0, 1) is missing, because when $x = 0, t = -\frac{1}{2}$, so $y = -1$; b. One-to-one

